Advanced Cloud Computing

ShopScale: E-commerce under real-world cloud workloads

Engineering Ideas into Real-World Solutions.



Cloud Computing DevOps Distributed Systems Cybersecurity AI Software Engineering

THE WORKSPACE

One workspace for building complete working systems

Build the whole system.

Acentralized GitHuborganizationfortechnologyprojects,practicalimplementations, experimentation, andtechnical learning.

Implement

Connect technologies into working applications.

Experiment

Test system behavior under controlled conditions.

Learn

Capture knowledge from practical engineering.

The emphasis is complete working systems, rather than technologies learned in isolation.

PROJECT OBJECTIVES

Four objectives drive every project

Build practical solutions

Develop software and technology that address real implementation needs.

Explore modern computing

Investigate cloud platforms and emerging computing technologies.

Evaluate system quality

Examine application performance, security, and reliability.

Document the evidence

Record technical knowledge and experimental outcomes.

ENGINEERING SCOPE

Six technical domains, one engineering practice



Cloud Computing and Distributed Systems
Scaling, load balancing, serverless



DevOps and Infrastructure Automation
CI/CD, containers, infrastructure as code



Performance Monitoring and Reliability Engineering
Metrics, dashboards, recovery testing



Cybersecurity and Information Security
Access control, least privilege



Artificial Intelligence and Data Engineering
Data pipelines, notebooks, analysis



Software and Application D evelopment
Web apps, APIs, databases

TECHNOLOGY TOOLKIT

A flexible stack, chosen per project

Programming
Python, JavaScript, SQL

Frontend
HTML, TailwindCSS

Backend
FastAPI, REST APIs

Cloud
AWS, Microsoft Azure, Google Cloud

D evOps
Docker, Jenkins, GitHub Actions, Terraform

Monitoring
Prometheus, Grafana, CloudWatch

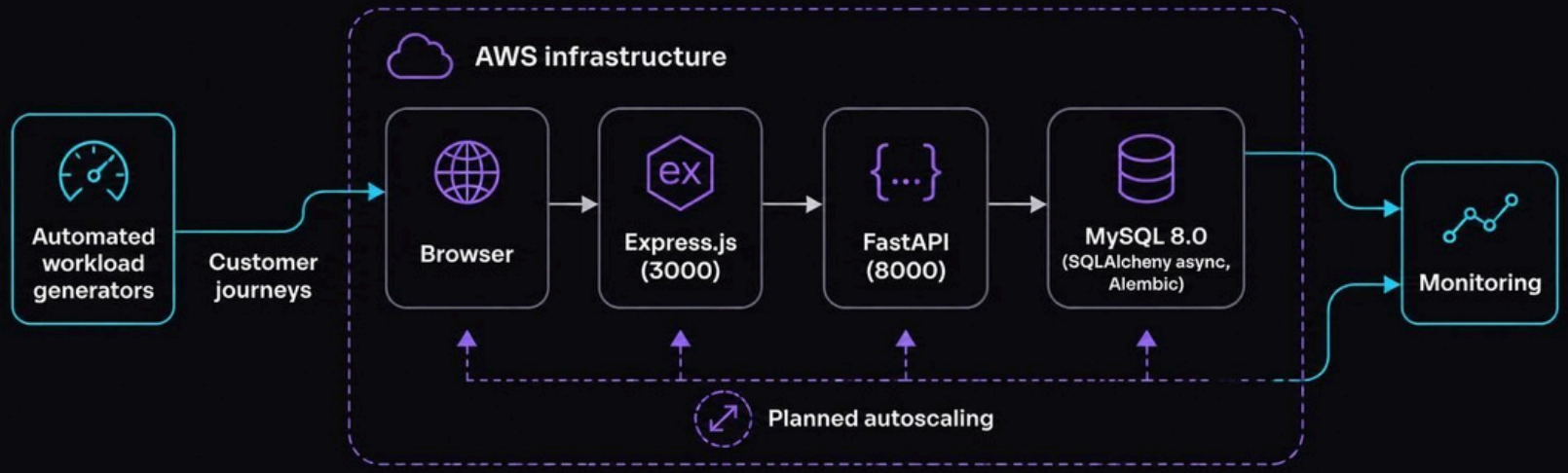
Databases
PostgreSQL, MySQL, Redis

Data Science
NumPy, pandas, Matplotlib, Jupyter

Technology choices vary by project.

ShopScale: planned e-commerce scalability and reliability

PLANNED / IN DEVELOPMENT NO COMPLETED DEPLOYMENT CLAIMED



Planned storefront

Product catalog, cart, inventory, simulated checkout

Planned experiments

Automated customer journeys, controlled traffic, performance monitoring

Planned evaluation

SLA evaluation, recovery testing, bot detection

Serverless and cloud-native research: areas of exploration

EXPLORATION · NOT VERIFIED IMPLEMENTATION



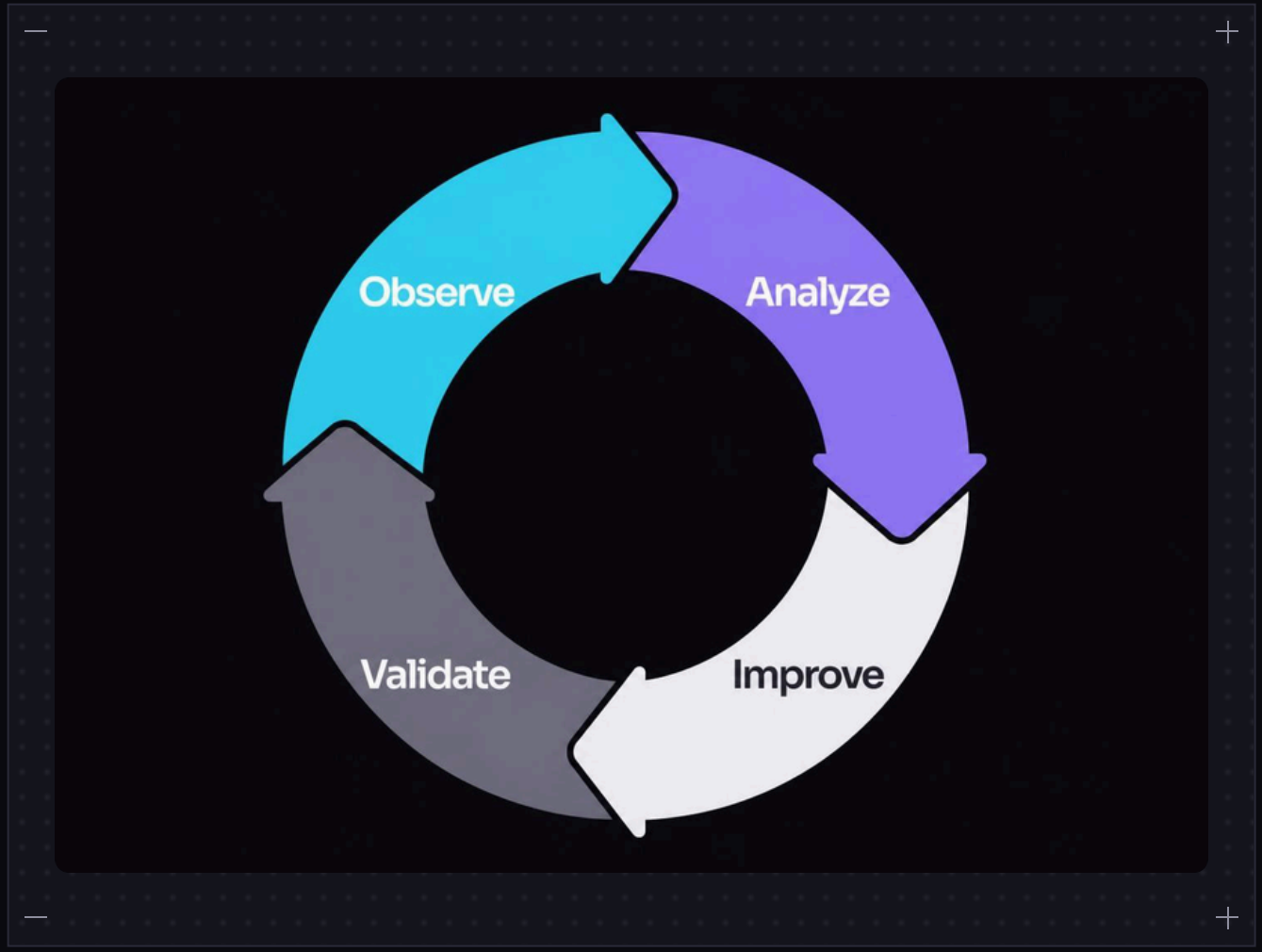
Compute
Serverless computing and Function-as-a-Service

Architecture
Distributed workflows, hybrid and multi-cloud architectures

Evaluation
Cloud storage and performance benchmarking

SYSTEM QUALITY

Dependable applications need observability and security



Obse r vability

Metrics, logging, resource utilization

Security

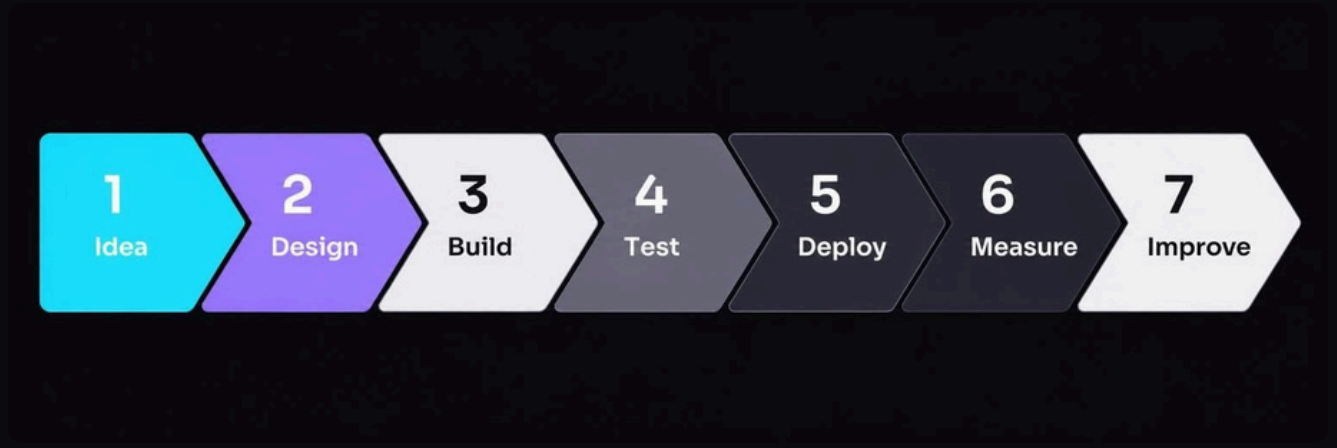
Access control, least privilege

Reliability

Performance testing, incident detection, recovery

ENGINEERING METHOD

Prototype first, then measure everything



- Build functional prototypes before adding complexity.
- Validate assumptions through controlled experiments.
- Measure outcomes instead of making unsupported claims.
- Document architecture, setup, testing, and limitations.
- Collaborate, review, and continuously improve.

Learn by Building.
Grow by Experimenting.

Ameaningful collectionofpracticalprojectsacrosscloudengineering,DevOps,distributed systems, AI, cybersecurity, andemergingtechnologies.

- From Technical Concepts to Practical Implementations.
- Practical projects. Controlled experiments. Documented learning.